Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Midrand

Faculty: Information Technology

Module Code: ITBDA

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Indicate	Yes	No
Plagiarism report attached	✓	

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature **Dian Kotze**	**Date** **2026/11/11**

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Question 1

Question 1.1

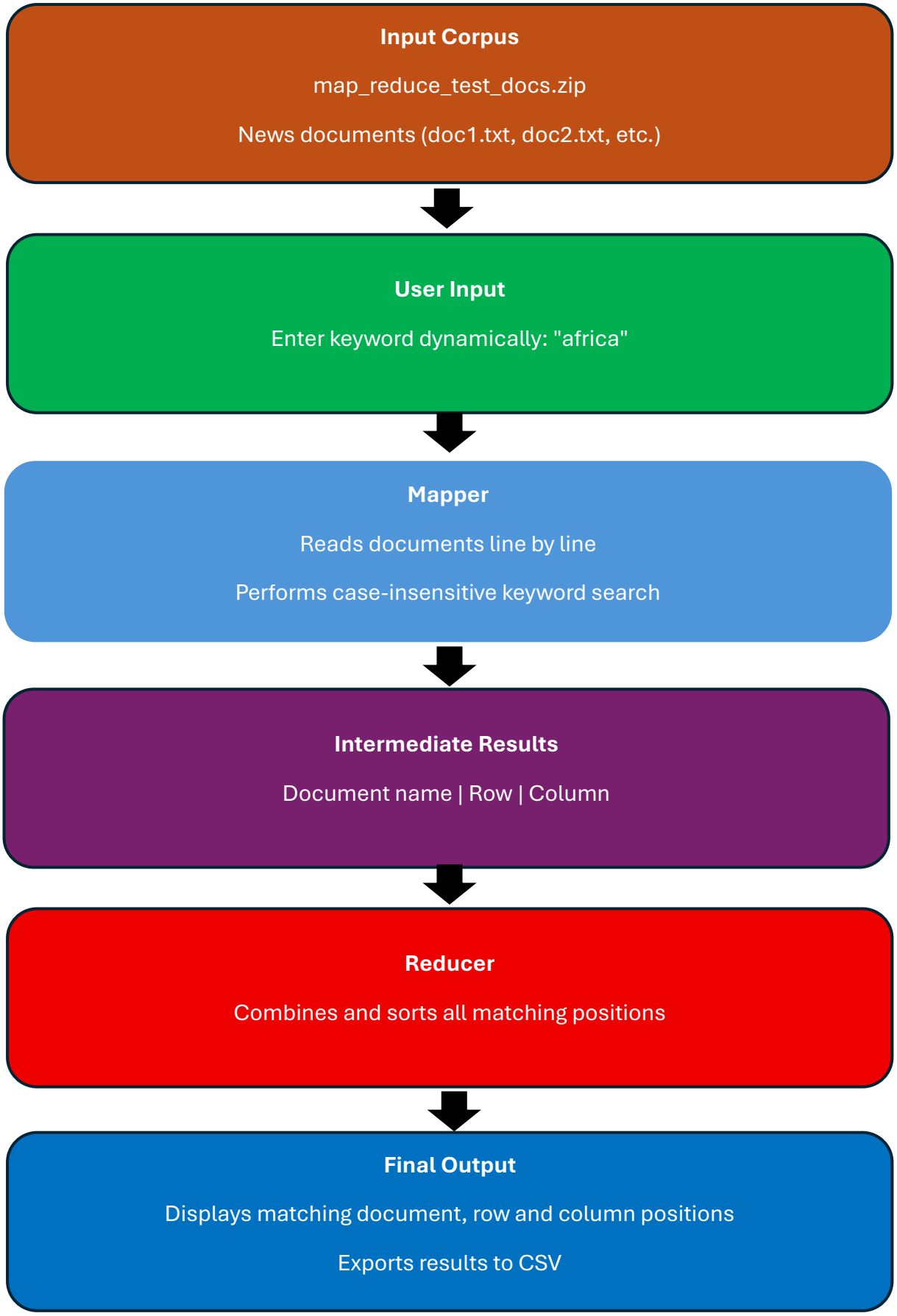
The MapReduce framework is used to process large amounts of data by dividing the task into smaller operations. For this project, the process is used to search for a specific keyword within a collection of news documents and identify the row and column position where the keyword appears.

The process starts with the input documents from the map_reduce_test_docs.zip file. These documents form the input corpus for the keyword search. The user enters a keyword dynamically, such as "africa", which is passed to the Mapper.

The Mapper reads each document line by line and searches for the keyword. The search is case-insensitive, so uppercase and lowercase versions of the keyword are treated equally. Whenever a match is found, the Mapper records the document name, row number and starting character position of the keyword.

The intermediate results are then passed to the Reducer. The Reducer combines the Mapper results and sorts them according to document name, row number and column position. Finally, the output displays all the matching locations and can also be exported to a CSV file.

This process makes it easier to search multiple documents and organise the results in a structured format.


Input Corpus
map_reduce_test_docs.zip
News documents (doc1.txt, doc2.txt, etc.)
User Input
Enter keyword dynamically: "africa"
Mapper
Reads documents line by line
Performs case-insensitive keyword search
Intermediate Results
Document name | Row | Column
Reducer
Combines and sorts all matching positions
Final Output
Displays matching document, row and column positions
Exports results to CSV

Explanation of the MapReduce Process

Step 1: Input data

The input consists of a collection of news article documents stored as text files. Each document is treated as a separate input record. The documents are loaded into R and assigned a document name.

Step 2: Mapper

The Mapper reads each document line by line. The user enters a keyword that must be searched for. The Mapper converts the text and keyword to lowercase to ensure that the search is case-insensitive.

When the keyword is found, the Mapper records the document name, row number and column position of the keyword. These results form the intermediate output.

Step 3: Shuffle and Sort

In a traditional Hadoop MapReduce environment, the framework groups and sorts the intermediate key-value pairs according to their keys. In this implementation, the results are organised according to the document and row number.

Step 4: Reducer

The Reducer receives the intermediate results and combines the occurrences of the keyword. It organises the results into a structured table containing the document name, row number, column number and keyword.

Step 5: Final output

The final output displays the location of every occurrence of the keyword in the documents. This makes it possible to identify which news articles contain the keyword and where it appears within each article.

The R implementation demonstrates the logical functionality of MapReduce. Although it does not execute on a Hadoop cluster, it provides a practical demonstration of the Mapper, Reducer and configuration logic.

Question 1.2

The Mapper is responsible for reading each document and searching for the specified keyword. In this project, the Mapper was implemented using R.

The keyword is entered dynamically by the user when the program runs, rather than being hardcoded into the script. The Mapper searches through each document line by line and compares the text with the keyword. The search is case-insensitive, meaning that uppercase and lowercase versions of the keyword are treated as the same word.

When the keyword is found, the Mapper records the document name, row number and column position where the keyword starts. These results are then passed to the Reducer for further processing.

The following R code demonstrates the Mapper implementation:

```
mapper <- function(documents, keyword) {
  map_results <- data.frame(
    document = character(),
    row = integer(),
    column = integer(),
    stringsAsFactors = FALSE
  )

  keyword <- tolower(keyword)

  for (doc_name in names(documents)) {
    lines <- documents[[doc_name]]

    for (row_number in seq_along(lines)) {
```

```
    line <- tolower(lines[row_number])

    positions <- gregexpr(keyword, line, fixed = TRUE)[[1]]

    if (positions[1] != -1) {
      for (column_position in positions) {
        map_results <- rbind(
          map_results,
          data.frame(
            document = doc_name,
            row = row_number,
            column = column_position
          )
        )
      }
    }
  }
}

  return(map_results)
}
```

The Mapper function receives the documents and the keyword as input. It converts both the keyword and document text to lowercase to ensure that the search is case-insensitive. It then uses nested loops to examine each line in every document. The

gregexpr() function identifies the starting character positions of all occurrences of the keyword. Each match is stored with its document name, row number and column position. The Mapper returns a data frame containing these results, which can then be processed by the Reducer.

Enter the keyword to search for: africa

--- MAPPER OUTPUT ---

	document	row	column	keyword
1	doc1.txt	1	64	africa
2	doc1.txt	3	11	africa
3	doc1.txt	3	95	africa
4	doc2.txt	1	57	africa
5	doc2.txt	3	32	africa
6	doc2.txt	33	24	africa
7	doc2.txt	58	5	africa
8	doc2.txt	58	81	africa
9	doc3.txt	1	38	africa
10	doc3.txt	3	7	africa
11	doc3.txt	5	7	africa
12	doc3.txt	40	146	africa
13	doc3.txt	46	7	africa
14	doc3.txt	48	112	africa
15	doc3.txt	50	142	africa
16	doc3.txt	52	86	africa
17	doc3.txt	54	44	africa

18 doc5.txt 1 62 africa

19 doc5.txt 3 88 africa

20 doc5.txt 23 121 africa

21 doc5.txt 38 27 africa

22 doc5.txt 48 217 africa

--- REDUCER OUTPUT ---

document row column keyword

1 doc1.txt 1 64 africa

2 doc1.txt 3 11 africa

3 doc1.txt 3 95 africa

4 doc2.txt 1 57 africa

5 doc2.txt 3 32 africa

6 doc2.txt 33 24 africa

7 doc2.txt 58 5 africa

8 doc2.txt 58 81 africa

9 doc3.txt 1 38 africa

10 doc3.txt 3 7 africa

11 doc3.txt 5 7 africa

12 doc3.txt 40 146 africa

13 doc3.txt 46 7 africa

14 doc3.txt 48 112 africa

15 doc3.txt 50 142 africa

16 doc3.txt 52 86 africa

```
17 doc3.txt  54     44  africa
18 doc5.txt   1     62  africa
19 doc5.txt   3     88  africa
20 doc5.txt  23    121  africa
21 doc5.txt  38     27  africa
22 doc5.txt  48    217  africa

--- MAPREDUCE CONFIGURATION ---
$framework
[1] "R MapReduce Simulation"

$input_format
[1] "Text Files"

$output_format
[1] "CSV"

$mapper
[1] "Keyword Mapper"

$reducer
[1] "Keyword Reducer"

$case_sensitive
```

[1] FALSE

$keyword

[1] "africa"

MapReduce process completed successfully.

Question 1.3

The Reducer is responsible for processing the intermediate output generated by the Mapper. Its purpose is to organise and combine the search results into a structured format.

In this implementation, the Reducer receives the data frame generated by the Mapper. It sorts the results according to the document name, row number and column number. This ensures that the output is organised and easy to interpret.

The Reducer also checks whether any keyword occurrences were found. If no matches are available, it returns an empty data frame instead of generating an error.

The final output contains the document name, row number, column number and searched keyword. This provides the required location information for every occurrence of the keyword.

The Reducer implementation is included in the R script under the reducer() function.

Question 1.4

The configuration settings define how the MapReduce process operates. These settings specify the input format, output format, Mapper function, Reducer function, case sensitivity and keyword used during the search.

The configuration is implemented using an R list. The keyword is assigned dynamically when the user enters it, which allows the same program to be used to search for different words without modifying the source code.

The following configuration settings are used:

Configuration	Description
Framework	R-based MapReduce simulation
Input format	Text files
Output format	CSV
Mapper	Searches for the keyword and identifies its positions
Reducer	Organises the Mapper output
Case sensitivity	Disabled
Keyword	Entered dynamically by the user

The configuration settings are implemented using the mapreduce_config list in the R script.

The program saves the final results in a CSV file named MapReduce_Keyword_Results.csv, which can be used for further analysis.

It is important to note that these settings simulate the MapReduce configuration in R. They are not native Hadoop configuration settings and do not connect the program to a Hadoop cluster.

Question 2

Question 2.1

Introduction

Text mining is the process of extracting useful information from unstructured text data. News articles contain large amounts of textual information, which can be analysed to identify common themes and topics.

For this practical, a collection of news articles is used as the input dataset. The articles are loaded into an R data frame containing two columns: file_name and content. The file name identifies the article, while the content column contains the complete text of the article.

Data Preprocessing

Before applying topic modelling, the text must be cleaned to remove information that does not contribute meaningfully to the analysis. The following preprocessing techniques are applied:

- Lowercase conversion: Converts all text to lowercase to avoid treating words with different capitalisation as separate terms.
- Removal of punctuation: Removes punctuation marks such as commas, full stops and quotation marks.
- Removal of numbers: Removes numerical values that are not relevant to the topic analysis.
- Removal of stop words: Removes common words such as "the", "is", "and" and "of".
- Whitespace removal: Removes unnecessary spaces between words.
- Tokenisation: Splits the cleaned text into individual words.
- Document-term matrix creation: Converts the processed text into a numerical representation suitable for LDA.

These preprocessing techniques improve the quality of the data and help the topic modelling algorithm identify meaningful words and patterns within the news articles.

Question 2.2

Latent Dirichlet Allocation (LDA) is an unsupervised machine learning technique used to identify hidden topics within a collection of documents. It assumes that each document contains a mixture of topics and that each topic is represented by a distribution of words.

For this practical, LDA is applied to the preprocessed news articles to identify the main topics discussed in the collection. The number of topics is selected as three for demonstration purposes.

The model produces two important distributions. The first is the topic-word distribution, which indicates the probability of a word belonging to a particular topic. The second is the document-topic distribution, which indicates the proportion of each topic within a document.

The top words for each topic are selected based on their word probabilities. These words are used to interpret and assign a descriptive label to each topic. For example, words such as "government", "election" and "political" may indicate a political topic, while words such as "team", "match" and "player" may indicate a sports topic.

The topic distribution results are reported using tables showing the top words and their probabilities. The document-topic distribution is also used to identify the dominant topic within each news article.

LDA is useful because it can identify common themes in a large collection of documents without requiring the articles to be manually classified beforehand.

Question 2.3

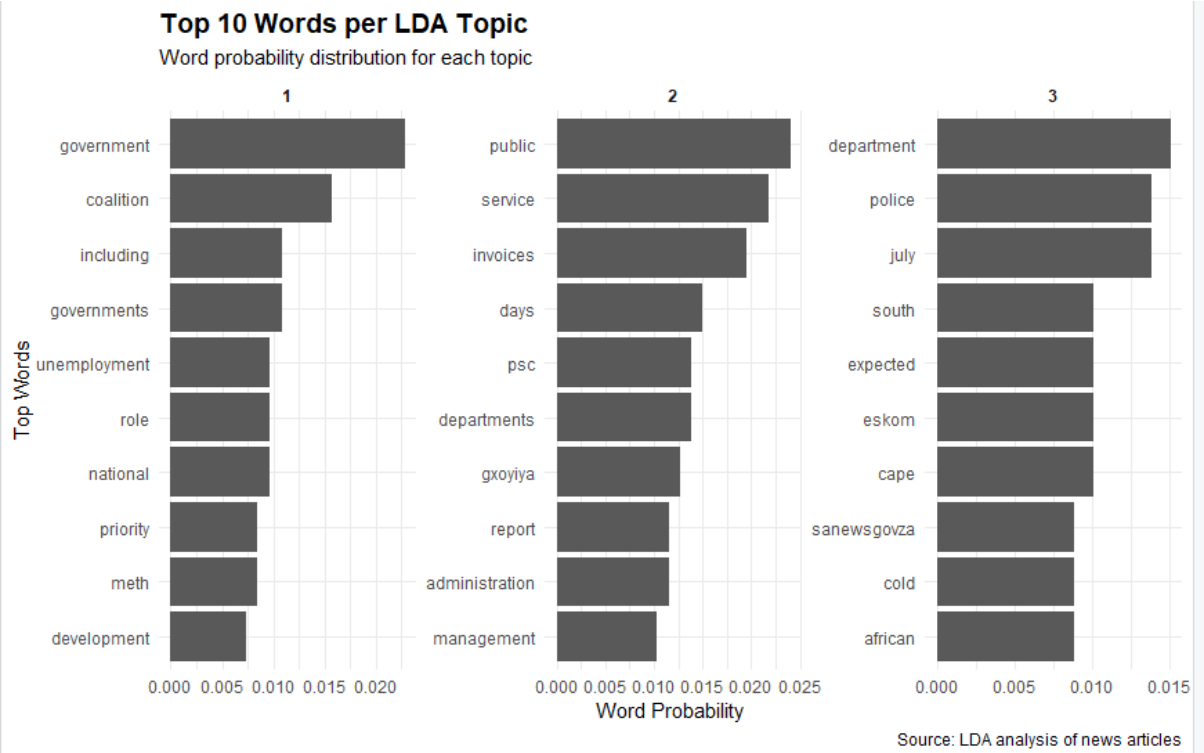
Visualisations are used to present the results of the LDA model in a clear and understandable manner. Two visualisation techniques are used in this practical.

The first visualisation is a faceted bar chart showing the top words for each topic according to their probability. This allows the most important words within each topic to be compared.

The second visualisation is a word cloud for each topic. The size of each word represents its importance within the topic. Words with higher probabilities appear larger than words with lower probabilities.

The visualisations assist in interpreting the topics identified by the LDA model. The bar chart provides a more precise comparison of word probabilities, while the word clouds provide a visual overview of the main terms associated with each topic.

Together, these visualisations make it easier to identify the main themes within the news article collection.



Question 3

Question 3.1

For this practical, the World Health Organization (WHO) website was selected as the source of information. WHO is an international organisation that provides information and resources relating to public health, diseases and healthcare.

The website contains publicly accessible information that can be used for educational and research purposes. A small selection of information pages was chosen for the practical. The R programming language was used to access the selected pages and extract their textual content.

The rvest and httr packages were used to retrieve the website content and extract the relevant paragraphs. The extracted information was cleaned by removing unnecessary characters, reference numbers and extra spaces.

The cleaned text was then stored in an R data frame. The data frame contains the page title, website URL and extracted text content. This structure makes the information easier to organise and allows it to be used in the following text summarisation and visualisation stages.

To follow responsible web scraping practices, only publicly accessible pages were selected, a limited number of requests were made, and a delay was included between requests. No login restrictions or security measures were bypassed.

CODE OUTPUT

--- MOST FREQUENT WORDS ---

	word	frequency
1	health	82
2	air	48
3	pollution	37
4	climate	36
5	food	26
6	global	26

7 physical 23

8 change 19

9 healthy 14

10 policies 14

11 activity 13

12 active 12

13 energy 11

14 people 11

15 can 10

--- FREQUENCY-BASED SUMMARIES ---

A tibble: 12 × 3

article_name sentence frequency_score

<chr> <chr> <dbl>

1 Air Pollution Episode 66 - Air pollution, a public health emergency Why does WHO consider air pol… 29.9

2 Air Pollution Learn about six air pollutants that are harming your health. 23.7

3 Air Pollution Episode 56 - Air pollution COVID-19 What have we learned about air pollution and CO… 23.4

4 Climate Change Global Action Plan on Climate Change and Health a strategic framework adopted by Me… 19.1

5 Climate Change This technical report presents a global assessment of the health, environmental and… 18.9

6 Climate Change WHO s work plan on climate change and health includes From commitments to implement… 15.9

7 Healthy Diet A healthy diet is a foundation for health, well-being, optimal growth and developme… 15.5

8 Healthy Diet The benefits of a healthy diet are reflected in higher educational outcomes, produc… 12.6

9 Healthy Diet WHO regularly monitors the adoption and implementation of food environment policies… 12.2

10 Physical Activity Physical inactivity remains a major global public health concern, with important im… 17.3

11 Physical Activity Sedentary behaviour and insufficient physical activity each contribute to poorer he… 12.6

12 Physical Activity It also helps to maintain a healthy body weight and can improve mental health, qual… 11.1

--- TF-IDF-BASED SUMMARIES ---

A tibble: 12 × 3

article_name sentence tfidf_score

<chr> *<chr>* *<dbl>*

1 Air Pollution If you live in highly polluted areas does COVID-19 affect you differently? 6.06

2 Air Pollution The WHO Ambient Air Quality Database compiles ground-level measurements of annual mean … 5.76

3 Air Pollution WHO is the custodian agency of a handful of SDG indicators for Goals 3 pertaining to go… 5.73

4 Climate Change In the longer term, the effects will increasingly depend on the extent to which transfo… 6.03

5 Climate Change Areas with weak health infrastructure mostly in developing countries will be the least … 6.03

6 Climate Change These climate-sensitive health risks are disproportionately felt by the most vulnerable… 5.77

7 Healthy Diet What about so called natural sweetness like Stevia? 6.46

8 Healthy Diet However, the basic principles of healthy diets remain the same for everyone. 5.90

9 Healthy Diet The chronic exposure to some of these substances... 5.86

10 Physical Activity These conditions are projected to cost approximately US 300 billion in healthcare expen… 5.86

11 Physical Activity The estimates do not include wider indirect costs, such as productivity losses. 5.75

12 Physical Activity Building policy to promote physical activity Ensuring access to sports for all Promotin… 5.72

Question 3.2

After collecting and cleaning the website content, two text summarisation techniques were applied using R. The purpose of these techniques was to identify important sentences from the four collected WHO pages and produce shorter summaries of their content.

Frequency-Based Summarisation

The first technique uses word frequency to identify important sentences. The program calculates how often meaningful words appear throughout the collected website content. Common English stopwords are removed so that words such as "the", "and" and "is" do not affect the results.

Each sentence is given a score based on the average frequency of its meaningful words. Sentences with higher scores are considered more important because they contain words that occur more frequently throughout the collected content.

The three highest-scoring sentences from each website page are selected to form the summaries.

TF-IDF-Based Summarisation

The second technique uses Term Frequency-Inverse Document Frequency (TF-IDF) to identify important sentences. TF-IDF measures the importance of words within a sentence compared with their occurrence across the entire collection of sentences.

Words that appear frequently within a sentence but are less common throughout the other sentences receive greater importance. Each sentence is given a score based on its average TF-IDF value.

The three highest-scoring sentences from each website page are then selected as summaries.

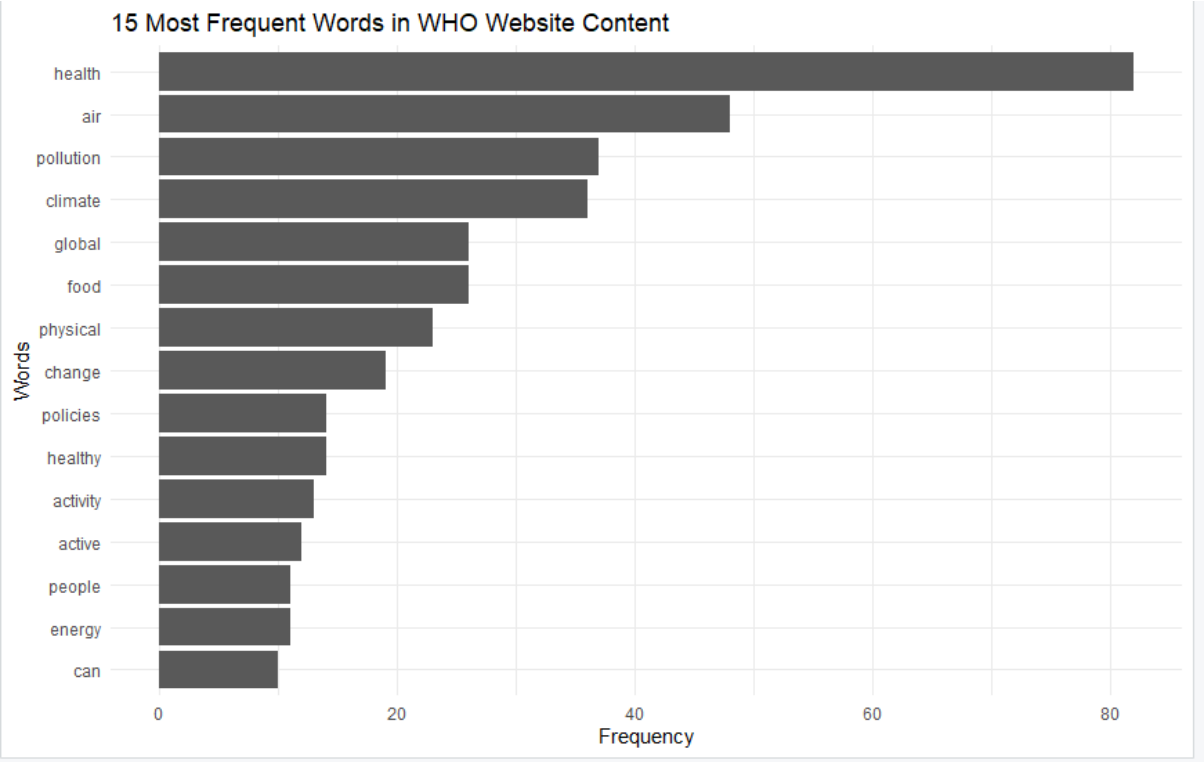
Comparison of Techniques

Both techniques are used to identify important information, but they approach the task differently. Frequency-based summarisation focuses on words that occur frequently throughout the collected content, while TF-IDF gives more importance to words that are distinctive within individual sentences.

Using both techniques provides two different ways of identifying relevant information and helps demonstrate how text summarisation can reduce large amounts of website content into shorter summaries.

Results of Text Summarisation

Both summarisation techniques were successfully applied to the four collected WHO pages. The frequency-based method and TF-IDF method each selected three sentences per page, producing twelve summary sentences for each technique. The results show that different summarisation methods can identify different types of important information from the same website content.



Question 3.3

Two visualisations were created using R to present the results of the text analysis.

Visualisation 1: Most Frequent Words

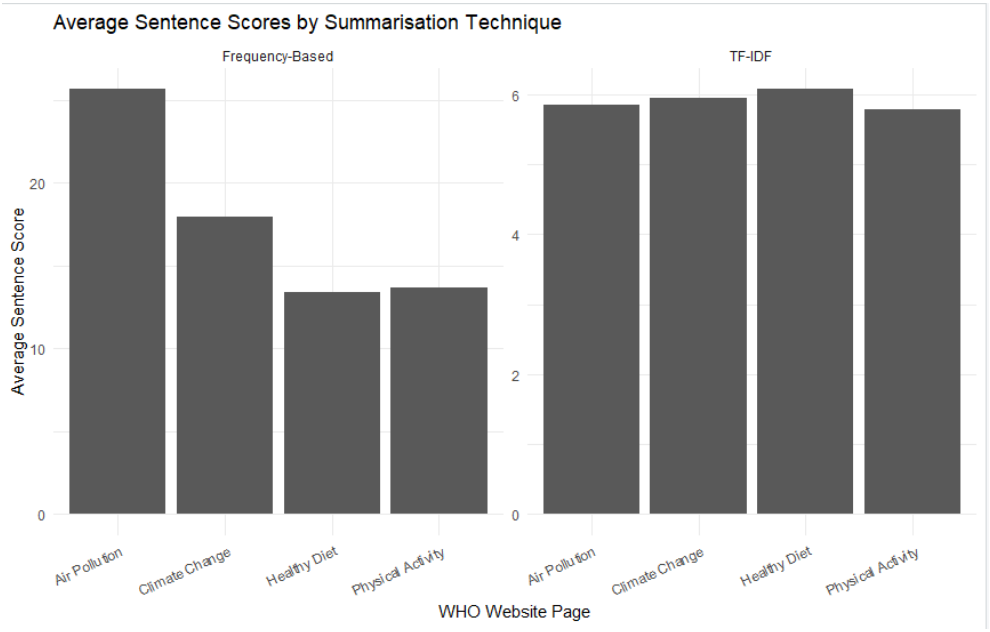
The first visualisation is a bar chart showing the 15 most frequently occurring words in the collected WHO website content. The chart displays each word and its frequency. The results show that words such as "health", "air", "pollution" and "climate" appeared frequently in the collected information. This helps identify the main subjects discussed across the selected website pages.

Visualisation 2: Comparison of Summarisation Techniques

The second visualisation compares the average sentence scores produced by the frequency-based and TF-IDF summarisation techniques for each WHO page. The chart shows the average scores of the three selected sentences for each page.

The visualisation uses separate scales for the two techniques because their scoring methods produce different numerical values. The results help show how each technique identifies and scores important sentences from the collected website content.

Overall, the visualisations make the text analysis results easier to understand. They provide a clear view of the most common words and show how the two summarisation techniques produced their results.



Question 4

Question 4.1

Hacker News was selected as the website for this practical because it provides publicly accessible technology news and discussion information. The website contains individual news stories with information such as article titles, links, authors, points, comments and posting times.

The R programming language was used to connect to the website and extract the information from its homepage. The rvest, httr and xml2 packages were used to retrieve and process the website content. Regular expressions were applied using the stringr package to identify and extract numerical information, including the number of points and comments associated with each story.

A total of 30 news records were collected. The extracted information was stored in an R data frame containing six attributes: title, link, author, points, comments and posting time. The data was then exported to a CSV file for further cleaning and analysis.

Only a single homepage request was made during the collection process. The information was collected for educational purposes without attempting to access restricted content.

Question 4.2

After collecting the Hacker News data, the dataset was inspected and cleaned using R. The initial dataset contained 30 records and six attributes: title, link, author, points, comments and posting time.

The dataset was checked for missing values and duplicate records. The inspection showed that there were no missing values and no duplicate records or titles. The author information was also checked and the extraction process was corrected to ensure that author names were collected successfully.

The data was then prepared by removing unnecessary spaces and ensuring that the points and comments attributes were stored as numerical values. Duplicate titles and incomplete records were also handled during the cleaning process.

After cleaning, the dataset still contained all 30 records and six attributes. No records needed to be removed because the collected data was complete. The final cleaned dataset was exported as Q4_Cleaned_Hacker_News_Data.csv for further analysis and database storage.

Question 4.3

After importing the cleaned dataset into PostgreSQL, two SQL queries were performed to investigate the Hacker News data.

Investigation 1: Identifying the Most Popular News Stories

The first SQL query was used to identify the ten news stories with the highest number of points. The results showed that "Foldable Flight: The Best Paper Airplanes" was the highest-ranked story, with 1,018 points and 151 comments. Other popular stories included the USPS ZIP Code boundary polygons story, with 617 points, and the Google Gemini story, with 595 points.

The results also showed that the story with the highest number of points did not necessarily have the highest number of comments. For example, "Claude Fable 5.2 and Mythos 5.2" received 790 comments, which was the highest comment count among the ten results.

Investigation 2: Relationship Between Points and Comments

The second SQL query calculated the correlation between the number of points and comments for the collected stories. The correlation result was 0.594, which indicates a moderate positive relationship between these two attributes.

This suggests that stories receiving more points generally also tend to receive more comments. However, the relationship is not perfect, meaning that some stories can receive a large number of comments without necessarily having the highest number of points.

Both investigations demonstrate how SQL can be used to retrieve and analyse information from a structured database.

Question 4.3

I created a GitHub repository, https://github.com/DKotze2018/ITBDA-Assignment-Midrand-EDUV4810649.git